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Moderation analysis in business and management research: Common issues, solutions, and guidelines for future research

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A R T I C L E I N F O

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A B S T R A C T

Moderation analysis is a critical part in business and management research, particularly within the information systems (IS) domain, yet it continues to face persistent methodological issues. These issues not only threaten the reliability of results but also hinder theoretical advancements. To address these challenges, our study undertakes a comprehensive examination of moderation analysis. We commence with a concise synthesis of its conceptual evolution by reviewing 30 foundational publications that have shaped its development. Subsequently, we categorize and clarify the core moderation models, including two-way, three-way interactions, and moderated mediation, highlighting their appropriate application contexts and corresponding analytical techniques. Building upon this foundational knowledge, we identify and detail 15 prevalent methodological issues in moderation research, assessing their contemporary prevalence through an empirical investigation of 274 articles published in top-tier IS journals over the past three years. To equip researchers with actionable guidance, we propose a state-of-the-art, stage-based framework that encompasses the entire research lifecycle—from initial preparation and hypothesis development through design planning, data collection, sophisticated analysis, rigorous interpretation, to transparent reporting. Our contributions are fourfold. First, we present contemporary empirical evidence on the persistence of historical issues and identify emerging trends in moderation research. Second, we offer a comprehensive, stage-based framework that transcends existing piecemeal recommendations, providing actionable support across the research lifecycle. Third, we consolidate theoretical insights by tracing the conceptual evolution of moderation analysis and systematically classifying major moderation models. Finally, we address the identified critical issues throughout the research process, equipping researchers with empirically validated status assessments and evidence-based solutions. Overall, our study enriches the understanding of moderation analysis and equips researchers, journal editors, and practitioners with a robust methodological roadmap for conducting rigorous and theoretically informed moderation research.

1. Introduction

Moderation describes the condition under which a causal relationship between variables varies across the values of one or more other variables (Baron & Kenny, 1986). The "other variables" affecting the direction and magnitude of the association are termed "moderators," "moderator variables," or "moderating variables". Moderation analysis is important for advancing theory in information systems (IS) and other business and management domains (e.g., Dong et al., 2025; Jo, 2025; Winterich et al., 2013), as it explains the specific conditions under which effects occur, leading to a more nuanced understanding of phenomena. It enables researchers to refine and extend existing theories by specifying the boundaries of theoretical relationships, fostering the development of more complex and accurate models. Incorporating moderating effects ensures conclusions are contextually accurate and reduces the risk of misapplication (Andersson et al., 2014). Moderation analysis is also meaningful for practices by providing insights into when and how specific strategies or interventions are more or less effective (Gonzalez-Mulé & Aguinis, 2018, p. 2247). Moderation analysis has evolved from a methodological tool to a fundamental component of research due to its critical role in theory and practice.

Despite its significance, moderation research is continuously plagued

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by some major issues (e.g., measurement errors and insufficient statistical power) (Becker et al., 2023; Dash & Paul, 2021; Memon et al., 2019). Landmark reviews—such as Aguinis et al.'s (2017) examination of 205 strategic management studies—reveal alarming error rates, where only 0.5 % of articles avoided all identified pitfalls. Similar deficiencies permeate other management domains such as IS, hospitality and organizational behavior (Carte & Russell, 2003; Memon et al., 2019; Rasoolimanesh et al., 2021). These flaws risk Type I/II errors, theoretical overgeneralization, and ineffective practical prescriptions (Dawson, 2014).

Critically, while prior work has highlighted common errors in moderation analysis (e.g., Aguinis et al., 2017; Carte & Russell, 2003; Memon et al., 2019), two gaps persist. On the one hand, there is lack of contemporary empirical validation. Existing evidence primarily predates recent methodological advances, leaving it uncertain whether these issues endure in current research. This gap necessitates an up-to-date empirical assessment of moderation research. On the other hand, few studies systematically organize and evaluate the potential issues across the entire research lifecycle. The improper and mistaken practices could occur at any stage, from preparation and articulating the theoretical rationale to planning a research design, analyzing the data, and drawing conclusions (Wu & Zumbo, 2008). Inappropriate methodological choices during early stages can have remarkable consequences for substantive conclusions. Therefore, a stage-based guideline is essential to demonstrate the state-of-the-art practices for distinctive moderation models (Bergh et al., 2022).

To address this issue, our study first synthesizes key developments in moderation analysis by examining 30 foundational publications that have shaped its evolution. We then delineate the nature, application contexts, and analytical techniques for major moderation models (e.g., two-way, three-way, and moderated mediation), empowering researchers to select their appropriate analytical approaches. Building on this foundation, the study identifies 15 prevalent methodological issues in moderation analysis. Unlike previous studies that primarily documented problems, we further examine their ongoing prevalence through empirical investigation. Through a systematic analysis of 274 moderation studies published in leading IS journals from 2022 to 2024, we quantify the contemporary manifestation of each problem. Furthermore, to provide researchers with actionable solutions, we provide stage-based best-practice recommendations that form a comprehensive roadmap, guiding them from initial research preparation and hypothesis development through design planning, data collection, sophisticated analysis, rigorous interpretation, to transparent reporting. Appendix B includes an empirical demonstration using structural equation modeling (SEM) for moderation analysis, illustrating the practical application of robust methodological principles. This integrated approach not only enriches the understanding of moderation analysis but also provides practical tools for its implementation.

While foundational studies (e.g., Aguinis et al., 2017; Carte & Russell, 2003; Memon et al., 2019; Rasoolimanesh et al., 2021) have systematically documented methodological challenges in moderation analysis, our research advances the field through four key contributions. First, we present contemporary empirical evidence through a systematic review of 274 articles published in leading IS journals (2022–2024). This analysis not only quantifies the persistence of 15 historical issues (e.g., 72 % prevalence of b_3 misinterpretation as effect size) but also identifies 5 previously common problems that have significantly decreased in recent literature. Second, we develop an integrated stage-based framework that transcends existing piecemeal recommendations. Our framework provides actionable guidance across the complete research lifecycle - from theory development to final reporting (see Table 2) - directly responding to Bergh et al.'s (2022) call for phase-specific methodological support. Third, we contribute to theoretical consolidation by synthesizing 30 seminal works to: (1) trace the conceptual evolution of moderation analysis, and (2) systematically classify major moderation models (including two-way, three-way interactions and moderated mediation) with their corresponding analytical approaches, thereby facilitating appropriate model selection. Finally, our study extends current literature by comprehensively addressing 15 critical issues throughout the research process. For each identified problem, we provide empirically validated current status assessments and evidence-based solutions contextualized within our stage-based framework. These collective contributions serve to connect historical methodological insights with contemporary empirical findings, ultimately promoting more rigorous and theoretically informed moderation research in IS and other business and management domains.

2. Evolution of moderation

Remarkable advances in moderation analysis have been observed during the past several decades. We review and summarise 30 key articles that contributed to the development and diffusion of moderation. The procedure by which we identified these articles and their contributions to the evolution of moderation is summarised in Appendix A. Fig. 1 visualizes these representatives.

The label "moderator variable" seems to have been used first by Saunders, (1955); (1956), although the existence of moderation has been noted and discussed since the middle of the last century (e.g., Frederiksen & Melville, 1954; Gaylord & Carroll, 1948). Saunders (1956) was also the first to describe the hierarchical moderated multiple regression (MMR) method, widely used to assess the moderating relationship. Subsequently, the interest has increased dramatically in identifying the fundamental assumption of moderation and moderators, demonstrating the validity of moderation analysis, and determining how moderation analysis can be applied in different research contexts. For instance, Sharma et al. (1981) presented a framework to identify the presence and type of moderator variables. These early efforts contributed to the inception and creation of moderation analysis.

Early fundamental studies facilitated the acceptance and application of moderation in business and management research, triggering further refinement and extension of moderation analysis. Busemeyer and Jones (1983) evaluated the validity of the observational method in conjunction with MMR for empirically testing moderation concerning the impacts of measurement level and measurement error. Since some early studies used the terms moderator and mediator interchangeably, Baron and Kenny (1986) elaborated on the differences between moderators and mediators. Their research made theorists and researchers aware of the importance of distinguishing moderators and mediators and provided a compendium for further research to consider them simultaneously or separately. Russell and Bobko (1992) examined the impact of scale (a continuous dependent-response scale *vs.* a discrete Likert-type scale) in moderation testing. They warned researchers to note the information loss caused by using relatively coarse Likert scales. Cortina (1993) raised concerns regarding multicollinearity and recommended including the squared terms as covariates when testing moderation. Aguinis (1995), Aguinis et al. (2005), and Shieh (2009) noticed the statistical power issue of the MMR approach. Similarly, Irwin and McClelland (2001) identified and clarified inappropriate practices of the MMR approach. Landis and Dunlap (2000) and Russell and Dean (2000) noted the impacts of variable ordering and variable transformation in examining moderation. Chin et al. (2003) examined the effect size of using PLS-SEM to analyze moderation. Building on the prior work concerning single moderators, Dawson and Richter (2006) probed the issues related to the three-way interactions in MMR. Based on these studies, Aguinis and Gottfredson (2010) discussed the misapplications of MMR and provided good practices for moderation analysis. Other researchers are concerned with the theorizing issue of moderation (Andersson et al., 2014), test procedure of moderation testing (e.g., Hayes & Matthes, 2009), categorical and continuous variables (e.g., Aguinis et al., 2001), differentiating and/or integrating moderation and mediation (e.g., Frazier et al., 2004; Preacher et al., 2007), moderation in two-instance repeated measure designs (e.g., Montoya, 2019), and

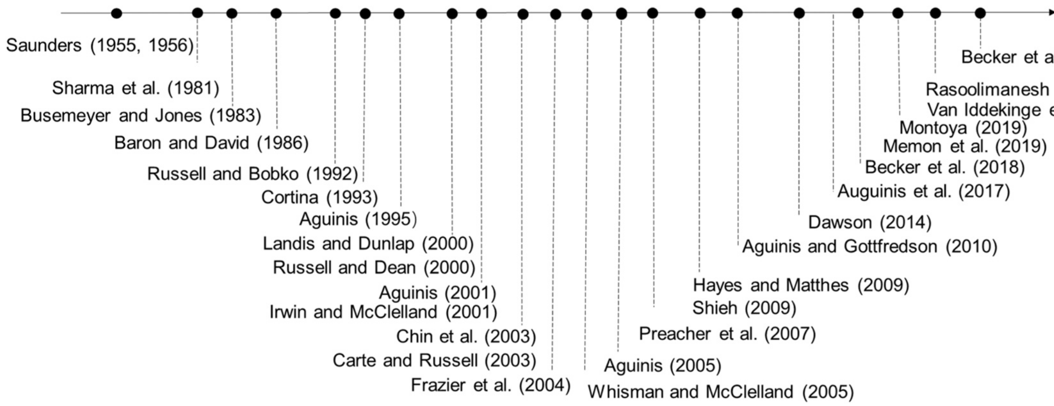

Fig. 1. Evolution of moderation analysis.

some other complex technologies (e.g., Dawson, 2014). These discussions not only demonstrated the need for moderation analysis but also prompted researchers to refine and extend the moderation analysis method based on its advantages and drawbacks.

Scholars' refinement and extension of moderation further contributed to disseminating moderation in various fields. Many IS and other management theories postulate moderation or interaction (Crook et al., 2008; Tang et al., 2024; Wang et al., 2023; Whisman & McClelland, 2005). Parallel to these applications, methodological research still observes problems and errors in moderation analysis. Therefore, to facilitate uniform application moderation analysis, some scholars provided user guides or best-practice articles to illustrate how moderation should be applied. For example, Carte and Russell (2003) provided guidelines to avoid nine common errors in pursuit of moderation based on a review of the MIS and broader management literature. Similarly, Aguinis et al. (2017) clarified the problems related to moderation from a review of the latest strategic management research. These studies contribute to moderating knowledge consolidation and dissemination. Becker et al. (2023) elaborated on many critical issues regarding moderation analysis using PLS-SEM. As nuances emerge with the development of IS and other management theories, scholars continue revisiting and clarifying taken-for-granted elements, investigating the boundary conditions, advantages, and limitations; identifying methodology gaps, and providing guidelines for moderation studies (e.g., Becker et al., 2018; Memon et al., 2019; Rasoolimanesh et al., 2021; Van Iddekinge et al., 2021). Such efforts provide substantial and continuous knowledge and practices for moderation research.

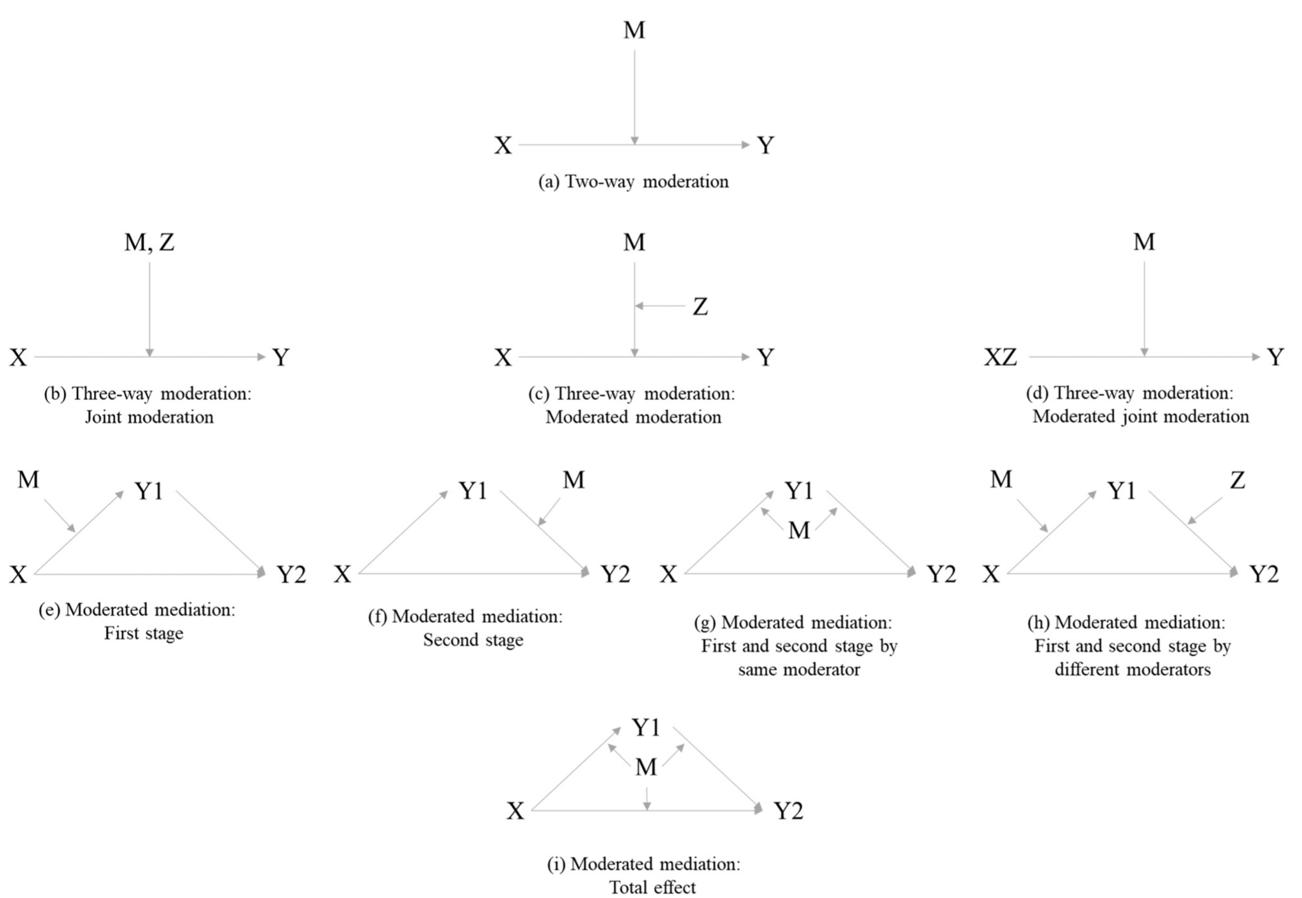

Fig. 2. Modes of moderation models.

3. Modes of moderation models

The growing application of second-generation multivariate data analysis techniques, such as PLS-SEM and CB-SEM (Hair et al., 2022; Hair & Alamer, 2022; Khan et al., 2019; Sarstedt et al., 2021; Sharma et al., 2024; Shiau et al., 2019; Shiau & Chau, 2016) has significantly expanded our capacity to test complex moderation models. Leveraging this methodological foundation of the identified core 30 seminal papers, we now examine several important and common moderation models, including two-way moderation, three-way moderation, and moderation models incorporating mediation.

3.1. Two-way moderation model

The two-way moderation model, illustrated graphically in Fig. 2(a), describes how the effect of an independent variable X on a dependent variable Y varies depending on a moderator M (Dawson, 2014). This model implies that M modifies both the strength and direction of the X-Y relationship, such as personal effort influencing the link between intelligence and academic performance. MMR is widely employed to test the two-way interaction. The core idea of MMR is to test H_0: $\Delta R^2 = R_2^2 - R_1^2 = 0$ using least squares procedures (ordinary or PLS-SEM), where

$$Y = \quad b_0 + b_1X + b_2M; R_1^2 \tag{1}$$

$$Y = \quad b_0 + b_1X + b_2M + b_3XM; R_2^2 \tag{2}$$

If ΔR^2 (i.e., the effect size) is significantly not equal to 0, it confirms the interaction's significance. If so, researchers often rewrite Eq. (2) to obtain the following equation

$$Y = (b_0 + b_2M) + (b_1 + b_3M)X \tag{3}$$

where $b_1 + b_3M$ is the simple slope, reflecting how the relationship between X and Y is moderated by the variable M. Furthermore, researchers are suggested to perform a simple slope test or at least moderation probe to evaluate if the form of the moderated relationship is consistent with their hypothesized direction.

Considering that M can be observed or latent, categorial (such as gender, race, education level, etc.) or continuous (such as age, years of education, number of stimuli, etc.), we then will elaborate on how to deal with different types M by the case of two-way moderation model. After that, we will also discuss how to probe two-way moderation.

3.1.1. Two-way moderation model analysis

3.1.1.1. Analysis of two-way moderation model with observed variables. (1) **When X and M are continuous variables,** the moderating effect can be analysed through MMR based on Eqs. 1 and 2. The moderation effect is significant when ΔR^2 is significantly not equal to 0. Additionally, it is generally believed that ΔR^2 exceeding 0.02 indicates a significant moderation effect with substantial meaning (Cohen et al., 2014). Still, even a small effect size may have practical significance if it supports the tested theoretical viewpoint, leads to severe consequences, or affects many individuals over time.

(2) **When X and/or M are categorical variables** (Y is a continuous variable), there are four different scenarios. Firstly, when both X and M are categorical, variance analysis can be conducted; η^2 or partial η^2 represents the moderating effective size, and the simple slope test is equivalent to the simple main effect test (Cohen et al., 2014). In the second scenario, one of X and M is a continuous variable, and the other is a dummy variable; MMR can still be used to analyze moderation (Hayes, 2018; Hayes et al., 2017). In the third scenario, M is a continuous variable while X is a categorical variable with k categories ($k \geq 3$). Researchers can encode X as a set of dummy variables and then perform MMR to analyze moderation. Taking $k = 3$ as an example: first of all, create two dummy variables X_1 and X_2 to represent X: $X_1 = X_2 = 0$ represent category 1, $X_1 = 1$ & $X_2 = 0$ represent category 2, and $X_1 = 0$ & $X_2 = 1$ represent category 3; second, let X_1, X_2, and M enter the regression model (R_1^2), then let all the product term (i.e., X_1M and X_2M) into the regression model (R_2^2); and finally, test the significance of ΔR^2. In the fourth scenario, X is continuous while M is categorical with k categories ($k \geq 3$); researchers can turn M into k dummy variables and follow the method mentioned above to examine the moderation or use grouped regression Analysis (GRA) to conduct moderation analysis. If GRA is employed, researchers can group the participants based on the categories of M, perform a linear regression analysis of X on Y within each group, and then test whether these regression coefficients are equal. If these coefficients are not all equal, it indicates that the moderation effect is significant. The drawback of GRA is that it can only detects moderation effect significance without quantifying effect size. Moreover, multiple tests increase the Type I error rate. A better approach is to conduct multigroup analysis (MGA) using SEM (see the following section).

3.1.1.2. Analysis of two-way moderation model with latent variables. (1) **When X is a latent variable, and M is a categorical variable**, MGA is recommended. MGA is an extension of GRA. The principle of MGA is based on comparing statistical models across different groups to determine whether the relationships between variables are consistent across these groups (Hair et al., 2022). Henseler et al. (2016) proposed a three-step measurement invariance assessment (MICOM) to perform MGA, which can be implemented using SmartPLS4 (the illustrative example in Appendix B provides details about how to implement MICOM). MGA also cannot estimate the effective size of the moderation effect.

(2) **When both the X and M are latent variables**, researchers can use MMR on factor scores or employ CB-SEM/PLS-SEM with product terms. Using MMR on factor scores is easy to understand, given that it involves factor analysis and regression analysis and can be completed using software like SPSS. However, given that this method uses factor scores and ignores the error, the standard errors of b_3 may be imprecise, affecting the test results. Therefore, we recommend using SEM with product terms, which has a theoretical foundation and software support, especially when the research model is relatively complex.

3.1.2. Two-way moderation model probe

When the moderation effect is significant, researchers need to further probe and visualize moderation to evaluate if the moderated relationship is consistent with their hypothesized direction. In particular, by providing graphs spanning the entire range of possible values of Y, it is possible to avoid exaggerating the size of the interaction and make it easier to accurately compare plots across studies even if they use different scales of measurement for Y (Murphy & Aguinis, 2022).

Two primary methods exist to probe an interaction: the pick-a-point approach (e.g., simple slope analysis, spotlight analysis) and the Johnson-Neyman (J-N) approach (e.g., floodlight analysis). The pick-a-point approach examines whether the effect of X on Y is significant at a specific value of M. The J-N approach generates the "region of significance" by identifying the critical values of M where the effect of X on Y shifts from significant to nonsignificant.

If the pick-a-point method is applied, researchers need first to select several specific points (commonly chosen as the mean, mean ± standard deviation (SD)) to represent M and then use the following t-statistic to test the significance of $b_1 + b_3M$.

$$t = \frac{b_1 + b_3M}{\sqrt{se_{b_1}^2 + 2M\text{cov}(b_1, b_3) + M^2 se_{b_3}^2}} \tag{4}$$

Here, se_{b_1} and se_{b_3} are the standard errors of b_1 and b_3, and $\text{cov}(b_1, b_3)$ is their covariance in Eq. (2). By plotting regression lines for Y on X at varying levels of M—specifically, high (mean + SD), medium (mean), and low (mean - SD), how the relationship between X and Y changes with the variation of M can be effectively illustrated. If the J-N method is applied, researchers need to use Eq. (4) to obtain two solutions $\widehat{M}$ (denoted as JM_{M_1} and JM_{M_2}, with $JM_{M_1} < JM_{M_1}$), such that when M equals $\widehat{M}$, the p value for the significant test of the simple slope $b_1 + b_3M$ is equals 0.05. Then, based on the value of JM_{M_1}, JM_{M_2}, $M_{\min}$, and $M_{\max}$ researchers can establish the range of M with which the simple slope $b_1 + b_3M$ significantly differs from 0. By plotting a graph with M as the horizontal axis and $b_1 + b_3M$ as the vertical axis at the 95 % confidence interval, it is possible to visually assess how the simple slope varies with the changes in M, and to identify the range of M that result in the simple slope being significantly non-zero (i.e., the confidence interval excludes 0). Both methods are supported by regression-based (e.g., PROCESS macro) and SEM-based (e.g., SmartPLS) tools. For details on the implementation of PLS-SEM, refer to Hayes (2018) and Ringle et al. (2015).

3.2. Three-way moderation models

When there are two or more moderators in the research models, researchers may consider using three-way interactions, that is, the product terms of the independent variable and multiple moderators, in their analytical procedures (Dawson & Richter, 2006). In three-way moderation, researchers need to explain how two moderators jointly affect the relationship between X and Y. We primarily consider three types of three-way moderation: joint moderation, moderated moderation, and moderated joint moderation.

3.2.1. Joint moderation model

In joint moderation, illustrated in Fig. 2(b), researchers need to explain why two variables would jointly moderate the relationship between X and Y (Lam et al., 2019). In this case, there is another moderator, W, that will also moderate the relationship between X and Y, in addition to M. Researchers should first depict the moderation effect of M and W respectively, and then discuss why the relationship of X and Y to be strongest/weakest under specific values of M and W. For instance, in a study examining the effect of implicit achievement motivation in shaping creative performance, Schoen (2015) proposed a joint moderation model to explain why individuals with high achievement motivation are more creative under conditions of high expected evaluation and domain knowledge level. Similarly, Baer (2012) proposed a three-way interaction model to explain how implementation instrumentality and strong buy-in ties jointly moderate the relationship between creativity and creative ideas implementation.

Joint moderation can be captured by the following regression model

$$Y = \quad b_0 + b_1X + b_2M + b_3W + b_4XM + b_5XW + b_6MW + b_7XMW \tag{5}$$

To test the joint moderation effect of M and W, researchers need first to run regression analysis to determine the three-way interaction term XMW is significant; Then, confirm the significance of the simple slopes in which the researcher expects the relationship to be positive (or negative) (e.g., high M and high W); Finally, test the significance of the simple slope differences (e.g., compare the slope of high M and high W against other three simple slopes) (Lam et al., 2019).

3.2.2. Moderated moderation model

In moderated moderation (i.e., Fig. 2(c)), the moderating effect of M is conditional on another moderator W (Lam et al., 2019). In other words, M moderates the relationship between X and Y in a specific condition (e.g., high W). In contrast, in another condition (e.g., low W), the moderating effect of M is insignificant. For instance, Arias-Pérez and Vélez-Jaramillo (2022) suggested a positive relation between digital orientation and digital innovation performance. They proposed that not-invented-here syndrome (NIHS) may weaken such a relationship. Given that employees begin perceiving new digital technologies as a threat to their careers, the proposed moderation effect of NIHS will be further moderated by employee's artificial intelligence and robotics awareness. Another example is from Stetz et al. (2006), which examined how social support's moderating effect on stressor-strain relationships depends on one's self-efficacy.

The moderated moderation regression model begins with confirming a significant XMW interaction, similar to joint moderation. However, the subsequent steps diverge due to the distinct conceptualization (Lam et al., 2019). In moderated moderation, the expectation is that the moderating effect of M will be evident specifically under conditions of high W. Therefore, it's necessary to demonstrate that the simple slopes at low and high M levels are statistically different (Dawson & Richter, 2006). After establishing that M significantly moderates the relationship at high levels of W, researchers must further validate that the nature of this moderation aligns with their theoretical expectations. This involves gathering empirical evidence to show that, with high W, the simple slope at low M levels is statistically significant. This step is crucial for confirming the specific form and direction of the moderated moderation effect.

3.2.3. Moderated joint model

In moderated joint moderation (i.e., Fig. 2(d)), the joint effect of XM is moderated by W. In this case, researchers are interested to know the boundary condition of the impact of XM on Y. For instance, Tepper et al. (2001) found subordinates' agreeableness (W) moderated how abusive supervision (X) and conscientiousness (M) jointly influenced constructive resistance (Y), emerging only under low agreeableness. Winterich et al. (2013) demonstrated that prosocial behavior recognition (W) moderated the interaction between moral identity symbolization (X) and internalization (M): symbolization increased prosocial behavior (Y) for low-internalization individuals only when recognition was present; without recognition, no interaction occurred.

To confirm the moderated joint moderation, researchers must conduct regression analysis and gather empirical data that satisfy three criteria. First, as with the above two types of three-way interactions, there must be a significant three-way interaction term XMW to support the hypothesis. Second, to show significant enhancing or substituting joint effects, the simple slopes at both low and high levels of M must be statistically distinct under the two conditions of W (i.e., when W is high and when W is low). Finally, the significance of simple slopes will vary based on whether the hypothesized joint effect is substituting or enhancing: For a substituting joint effect, both simple slopes (at high and low M) must be statistically significant at the theorized condition of W (whether that condition is high or low); For an enhancing joint effect, only the simple slope at high M needs to be statistically significant.

3.3. Moderation models with mediators

Mediation and moderation can be integrated through two approaches: moderated mediation and mediated moderation. Moderated mediation involves a conditional indirect effect where a moderator M interacts with a mediator Y1, altering the strength of the indirect effect of X on Y2 depending on M's value (Hair et al., 2022; Preacher et al., 2007). Specifically, this occurs when the mechanism linking X to Y2 via Y1 depends on M. Mediated moderation, conversely, refers to a moderating effect XM influencing Y2 indirectly through a mediator Y1, where Y1 explains the process underlying the moderation (Hair et al., 2022). However, since mediated moderation tests only an indirect effect of an interaction term without clarifying causal pathways, Hayes (2018) argues it offers limited analytical value and recommends prioritizing moderated mediation.

In moderated mediation models, moderation can take place in the first stage of the mediation effect (i.e., Fig. 2(e)), the second stage of the mediation effect (i.e., Fig. 2(f)), or both stages (i.e., Fig. 2(g) and (h)). In addition to the basic forms of moderated mediation, researchers may also consider the moderation of M on the direct effect of X on Y2 (i.e.,) (Edwards & Lambert, 2007). For instance, when investigating the effect of advertising irritation on digital advertising effectiveness, Sharma et al. (2022) not only examined the direct moderating effect of irritation on the relationship between advertising value (ADV) and purchase intention (PI) but also verified how ADV affect PI through attitude toward advertising subject to the moderation effect of irritation. Since this paper primarily focuses on moderation analysis, researchers should refer to Ng et al. (2024) to comprehensively understand moderated mediation analysis.

4. Common issues of moderation analysis in current research

With the nuanced developments in moderation analysis, a need arises to continue identifying and describing the moderation issues, characterizing their manifestations so that state-of-the-art solutions can be provided to mitigate them. Building upon our previous analysis of the 30 seminal papers, we identified 19 classic methodological issues (see column 1 of Table 1) historically associated with moderation analysis. To assess whether these issues persist in contemporary research and to document their current manifestations, we conducted a systematic diagnostic review of recent moderation studies published in three top IS journals. Our empirical investigation reveals that while 4 of the original 19 issues have been substantially mitigated through methodological advances, 15 critical challenges remain pervasive. These persistent issues demand urgent scholarly attention due to their continued impact on research validity. In this section, we first detail our diagnostic review. We then present and systematically analyze these 15 prevalent methodological issues, which are crucial for researchers to address when designing moderation studies and for reviewers to scrutinize during evaluation. Grounding our recommendations in these diagnostic insights, the subsequent section advances stage-specific best practices to enhance methodological rigor in moderation analysis.

4.1. Diagnostic review and findings

To diagnose contemporary methodological practices if moderation research, we executed a rigorous, multi-phase review process of research articles published from 2022 to 2024 in three premier IS journals: *MIS Quarterly* (MISQ), *Information Systems Research* (ISR), and *Journal of Management Information Systems* (JMIS). Our sampling frame comprised "Research Articles" and "Research Notes", excluding non-research content (e.g., editorials, commentaries). This yielded 580 candidate articles (205 from MISQ, 254 from ISR, and 121 from JMIS). Subsequently, we used the keywords “interaction”, “interacting”, “moderation”, and “moderating” to screen moderation research following Aguinis et al. (2017) and identified 319 potentially relevant articles. We then manually examined these articles to confirm 274 articles containing explicit moderation tests as our analytical sample. This resulted in 92 articles from MISQ, 119 articles from ISR, and 63 articles from JMIS. Each of the 274 articles underwent systematic manual coding to detect occurrences of the 19 predefined methodological issues, with results summarized in Table 1.

Our systematic review reveals an overall decline in the prevalence of methodological issues in moderation analysis. Notably, 4 historically documented problems (e.g., confounding XM with X^2, misuse of moderated mediation model, assuming the variable distributions include the full range of possible values, and the dependent variable scale is too coarse) exhibited markedly low incidence rates (<8 %). However, 15 problems persisted at clinically significant levels, among which 12 exhibited occurrence rates exceeding 10 % in contemporary research. In particular, 2 problems demonstrated particularly alarming pervasiveness: “lack of attention to the moderation effect size (71.53 % prevalence)” and “lack of attention to the moderation probe (51.83 % prevalence)”. We focus on the 15 high-impact, prevalent issues demanding further scholarly intervention—issues that will structure our subsequent diagnostic framework.

4.2. Common issues in current moderation analysis

For each of these 15 persistent issues, we clarify their conceptual nature, practical consequences, and evidence-based solutions. These issues are categorized into four parts, i.e., "model design and theorizing," "measurement, sampling, and scaling," "evaluation and analysis," and "interpretation and reporting". Such categorization not only identifies problems across the entire research continuum but also establishes the structural foundation for developing stage-based recommendations in the subsequent section.

4.2.1. Moderation model design and theorizing issues

Issue 1: Using moderation analysis driven by problematic motives

This widespread issue, identified in studies such as Dawson (2014), Memon et al. (2019), and Becker et al. (2023), pertains to the inappropriate justification for employing moderation analysis. Determining whether a moderation model is theoretically warranted constitutes the crucial first step in such studies (Memon et al., 2019). Despite increasing attention to moderation analysis, our review revealed that 29.2 % of the examined articles were driven by questionable assumptions. For instance, some studies use “trial and error” to identify possible moderators. This post-hoc strategy, lacking adequate theoretical grounding, is likely to produce meaningless conclusions even when statistically significant moderating effects are found. Moderation studies, which typically rely on null hypothesis significance testing (NHST), must instead be motivated by the research’s actual needs and underpinned by solid theoretical frameworks (Frazier et al., 2004). For example, moderation models are appropriately used to explain unexpectedly weak or inconsistent relationships between a predictor and an outcome across studies, or to test novel theoretical propositions (Andersson et al., 2014). Consequently, researchers should explicitly provide theoretical rationale—such as citing inconsistent prior findings or relevant contextual factors—for their proposed moderators (Memon et al., 2019).

Issue 2: Insufficient theoretical justification for moderation hypotheses

This persistent methodological issue, documented by Carte and Russell (2003), Dawson (2014), Landis and Dunlap (2000), and Rasoolimanesh et al. (2021), manifests primarily through insufficient theoretical justification for moderation hypotheses. The severity of this issue is underscored by Rasoolimanesh et al. (2021), whose analysis of 600 moderation studies revealed that only 25.61 % provided theoretical rationales for their moderation hypotheses, while 74.39 % failed to do so. Our review indicates significant improvement regarding this issue in IS research; nevertheless, it reveals that 32.85 % of studies either failed to articulate clear causal arguments or neglected to rationalize proposed moderation relationships. The lack of robust theoretical justification can lead to a series of adverse consequences. For instance, causal direction misspecification may yield divergent moderation outcomes, as reversing the X→Y and Y→X causal sequences—particularly with cross-sectional data—can alter statistical significance, risking post hoc theorizing absent causal justification (Carte & Russell, 2003). To mitigate these issues, it is theoretically necessary to prespecify causal paths and variable roles using empirical and/or theoretical evidence (Dawson, 2014). Where feasible, supplementing with panel data is also recommended. When ambiguity persists, conducting exploratory tests of alternative causal directions and variable designations, augmented by moderation probes (Hayes, 2018), can provide a more comprehensive understanding.

Issue 3: Failure to predefine hypotheses and their functional forms

Moderation analyses employing NHST require a priori hypothesis specification, given that NHST fundamentally evaluates predetermined expectations about moderating effects (Becker et al., 2023; Dawson, 2014; Rasoolimanesh et al., 2021). Our systematic review, however, identified that 34.67 % of studies either entirely omitted hypotheses or

Table 1
Common issues of moderation analysis.

Problems	Interpretations	Aguinis et al. (2017)	Rasoolimanesh et al. (2021)	This study	References
Moderation model design and theorizing issues					
Using moderation analysis driven by problematic motives	Investigating moderation with inappropriate reasons (e.g., merely doing it as others do or making the study complex) may result in spurious conclusions. Moderation analysis should be driven by actual theoretical needs.	X	X	29.20 % (n = 80)	Andersson et al. (2014), Dawson (2014), Frazier et al. (2004), Memon et al. (2019), Rasoolimanesh et al. (2021), Becker et al. (2023)
Insufficient theoretical justification for moderation hypotheses	The lack of robust theorizing can lead to a series of adverse consequences. It is recommended to prespecify causal and moderating paths supplemented with panel data verification.	X	74.39 %	32.85 % (n = 90)	Carte and Russell (2003), Dawson (2014), Glaser and Strauss (2017), Landis and Dunlap (2000), Rasoolimanesh et al. (2021)
Confounding X^2 and XM	The curvilinear relationship of X and Y may be recognized as moderation when X and M have high multicollinearity. Researchers are advised to examine the curvilinear effects for highly correlated X and M.	X	X	2.56 % (n = 7)	Cortina (1993), Dawson (2014), Hair et al. (2018)
Misuse of the moderated mediation model	Given that mediated moderation provides no additional insights into the path model effects, researchers are advised to disregard the concept of mediated moderation and focus on moderated mediation.	X	X	0.00 % (n = 0)	Aguinis et al. (2017), Dawson (2014), Hair, and Hult, Ringle, et al. (2022), Hayes (2015)
Failure to predefine hypotheses and their functional forms	Given that the moderation test relies on NHST, hypotheses specifying the impact and the scope of the moderating effects should be proposed in advance.	X	5 %	34.67 % (n = 95)	Becker et al. (2023), Dawson (2014), Mertens and Recker (2020), Rasoolimanesh et al. (2021)
Measurement, sampling, and scaling issues					
Lack of attention to measurement errors of the interaction term	Measurement errors weaken moderation analysis by distorting results. Despite growing attention, interaction terms' errors are often overlooked. Researchers must: use reliable measures, report reliability, and apply error-correcting methods like CB-SEM/PLS-SEM.	62.44 %	X	10.95 % (n = 30)	Aguinis et al. (2017), Carte and Russell (2003), Dawson (2014), Memon et al. (2019)
Assuming the variable distributions include the full range of possible values	Research samples are likely to represent a partial range of possible values on the variables in the population. Researchers should try to capture the full range of values of the variables when collecting data or at least provide the estimated population variance to rule out the range restriction effect.	34.15 %	X	7.30 % (n = 20)	Aguinis et al. (2017)
Nonlinear monotonic transformations on X, M, and Y without rational justification	Nonlinear monotonic transformations on X, M, and Y usually lack theoretical rationale and even generate some unintended consequences. If lacking a good reason for the transformation, researchers are suggested to use bootstrapping to estimate, given its ease of creation, the confidence interval of ΔR^2.	X	X	13.50 % (n = 37)	Busemeyer and Jones (1983), Carte and Russell (2003), Dawson (2014), Russell and Dean (2000)
Dependent variable scale is too coarse	In survey studies, the Likert scale, such as 5-point, 7-point, or 10-point, is frequently used to measure M. Given that each scale offers a unique set of advantages and limitations, the selection of scale should be based on a thorough evaluation of research objectives, information loss and statistical power, respondent characteristics, and previous research and conventions, with particular attention to balancing data precision and response rate.	X	X	3.65 % (n = 10)	Carte and Russell (2003), Dawes (2008), de Rezende and de Medeiros (2022), Russell and Bobko (1992)
Evaluation and analysis issues					
Low statistical power	The statistical power of the moderation test is generally lower than the main effect test, given the existence of measurement errors, random effect design, range restriction, scale coarseness, and other factors. A larger and more appropriate sample size and reliable measures can address this problem.	43.41 %	X	15.33 % (n = 42)	Aguinis et al. (2017), Carte and Russell (2003), Chin et al. (2003), Dawson (2014), Memon et al. (2019)
Converting a continuous variable into categorical without appropriate justification	Categorizing the moderators into dichotomous variables is highly associated with information loss and/or heteroscedasticity and may cause theoretical questions. Researchers should consider stopping the dichotomization if they cannot provide an appropriate justification	10.24 %	28.50 %	10.95 % (n = 30)	Aguinis et al. (2017), Becker et al. (2023), Carte and Russell (2003), Cohen et al. (2014), Dawson (2014), Memon et al. (2019), Rasoolimanesh et al. (2021)
Lack of attention to measurement invariance across categories	The testing for categorical moderators can be unreliable due to data inequivalence. Researchers are advised to collect similar numbers of samples for each category or perform measurement invariance before the MGA test.	20 %	63.90 %	13.50 % (n = 37)	Aguinis et al. (2017), Becker et al. (2023), Carte and Russell (2003), Hair, and Hult, Ringle, et al. (2022), Memon et al. (2019), Rasoolimanesh et al. (2021)

(*continued on next page*)

Table 1 (*continued*)

Problems	Interpretations	Aguinis et al. (2017)	Rasoolimanesh et al. (2021)	This study	References
Opaque or incorrect interaction term creation with SEM method	Latent interaction term creation faces persistent transparency/accuracy challenges, with widespread underreporting (73 % CB-SEM, 50 % PLS-SEM) and suboptimal manual methods. SEM offers three approaches—product indicator (collinearity-prone), orthogonalization (collinearity-resistant), and two-stage (formative-construct-required)—each requiring context-aware selection based on measurement type and research priorities	X	10.33 %	9.85 % (n = 27)	Becker et al. (2023), Rasoolimanesh et al. (2021)
Debate on mean-center/ standardize X and M	In general, mean-centering/standardization is recommended for the continuous moderating model, given that it can facilitate the interpretation of the coefficients without affecting the moderation tests; for the binary moderating model, researchers can consider leaving the variables in their raw form.	X	86.17 %	42.70 % (n = 117)	Aguinis et al. (2017), Becker et al. (2023), Dawson (2014), Hair & Hult, Ringle, et al., (2018), (2022), Memon et al. (2019), Rasoolimanesh et al. (2021)
Lack attention of lower-order interactions in three-way interaction model	Researchers often neglect lower-order interaction terms (XM/XW) when testing three-way interactions (XMW), risking misinterpretation. Rigorous model specification with all constituent terms and specialized software (e.g., SmartPLS) is critical for valid multi-moderator analyses.	X	X	5.11 % (n = 14)	Becker et al. (2023)
Interpretation and reporting issues					
Lack of attention of moderation effect size	The b_3 is not an appropriate indicator of the moderation effect, given that the ΔR^2 and b_3 are equal only when the product term is measured without error, and the variance of Y is equal to the variance of the product term. Researchers should use ΔR^2 or f^2 to conclude the moderation effect size.	X	> 40.5 %	71.53 % (n = 196)	Aiken et al. (1991), Becker et al. (2023), Carte and Rusell (2003), Chin et al. (2003), Hair, and Hult, Ringle, et al. (2022), Rasoolimanesh et al. (2021)
Ignoring nested model difference when interpreting main effect	If a moderating effect is significant, X has a range of effects on Y varying the level of M. Therefore, the main effects should be interpreted based on full models, including the product term.	42.93 %		29.20 % (n = 80)	Aguinis et al. (2017), Aiken et al. (1991), Becker et al. (2023), Dawson (2014), Memon et al. (2019), Rasoolimanesh et al. (2021)
Interpreting moderation based on the moderated model, excluding X and M	Excluding the model's predictors and/or moderators could confound the moderation effect with the main effect. Thus, researchers should include all predictors and moderators in the moderated regression equation.	X	X	26.64 % (n = 73)	Becker et al. (2023), Dawson (2014), Irwin and McClelland (2001)
Lack of attention to the moderation probe	Many existing moderation studies lack the necessary attention to moderation probes. However, it enables a clear understanding of how M influences the direction and/or strength of the relationship between X and Y. Researchers should use the pick-a-point or the J–N approach to visualize and probe moderation.	X	36.33 %	51.83 % (n = 142)	Becker et al. (2023), Dawson (2014), Hayes (2018), Memon et al. (2019), Murphy and Aguinis (2022), Rasoolimanesh et al. (2021), Ringle et al. (2015)

Note: This table compares the focus and findings of our study with prior methodological reviews. Our analysis is based on a sample of 274 articles published between 2022 and 2024 in the top three IS journals (MISQ, ISR, and JMIS). The percentages reported are based on this sample (N = 274).

presented incomplete specifications—a striking contrast to Rasoolimanesh et al.'s (2021) finding that only 5 % of studies failed to formally predefine hypotheses. This oversight may risk theoretical rigor and undermine moderation effects. Conversely, researchers should correctly hypothesize how the moderator alters the X→Y relationship—including functional form (e.g., strength amplification/diminution or direction reversal) and effect scope (e.g., full/partial model applicability)—prior to empirical testing, barring explicitly exploratory/post-hoc designs. Especially when the moderator is categorical, simply stating that "M (positively or negatively) moderates the effect of X on Y" is confusing. A properly specified hypothesis from our illustrative study demonstrates this rigor: "the effect of financial support on consumers' affective commitment to the streamer is stronger for females than males under the full model." Consistent with Issue 2, all hypothesized moderation mechanisms demand explicit theoretical justification prior to testing, except in explicitly exploratory or post-hoc analyses.

4.2.2. *Measurement, sampling, and scaling issues*

Issue 4: Lack of attention to measurement errors of the interaction term

Measurement errors constitute a primary source of low statistical power in moderation analysis, biasing unstandardized coefficient estimates when predictors X, M are measured with error—directly undermining the significance of ΔR^2—and attenuating explained variance estimates when the outcome Y is measured with error (Aguinis, 2017; Carte and Russell; Dawson, 2014; Memon et al., 2019). While attention to this issue in top journals appears to have increased (evidenced by a reduction in neglect, estimated at 10.95 %, in our review), potentially facilitated by greater access to secondary data for robust analysis, a significant problem persists: more than half studies employing latent variables still neglect the specific measurement error inherent in interaction terms. Therefore, despite some progress, the rigorous handling of measurement errors, particularly for interaction terms within latent variable frameworks, remains an essential requirement to avoid falsely attributing insignificant moderating effects to low true relationships rather than methodological limitations. Therefore, researchers need to use reliable measures and report reliability estimates to prove the reliability of their measures so that they exclude insignificance moderating effects caused by large measurement errors (Aguinis & Edwards, 2014; Aguinis & Vandenberg, 2014). The approach to generating the reliability of the product term can be found in Busemeyer and Jones (1983). Another effective solution to address measurement errors is to use CB-SEM or PLS-SEM with latent variables, given that SEM "offers important advantages over procedures that ignore measurement errors completely" (Aguinis et al., 2017, p. 678).

Issue 5: Nonlinear monotonic transformations on X, M, and Y without rational justification

Moderation analyses frequently require satisfaction of parametric assumptions, including multivariate normality of predictors (X and M) and homoscedasticity of prediction errors (Carte & Russell, 2003). Nonlinear monotonic transformations (e.g., log, square root, arc-sine) are commonly employed to achieve normality. 13.50 % of moderation studies in our review. The primary concerns with arbitrary transformations are threefold. First, they often lack theoretical justification, potentially distorting the true underlying relationships (Busemeyer & Jones, 1983). Second, transformed variables may yield unreliable MMR results, particularly when applied to outcome variables (Y). Third, transformations can artificially inflate Type I error rates or obscure genuine interaction effects. Addressing these problems requires a multifaceted approach centered on methodological robustness. Bootstrap methodologies (Russell & Dean, 2000) offer a viable alternative for estimating ΔR^2 confidence intervals without imposing strict distributional requirements. Comprehensive diagnostic testing of parametric assumptions should precede any consideration of variable transformation, with particular attention to Q-Q plots and heteroscedasticity tests. In cases where transformations are theoretically justified, researchers must provide explicit documentation of both the mathematical transformation and its substantive rationale. For severely non-normal distributions, robust regression techniques or permutation-based approaches may provide more reliable alternatives to traditional parametric methods.

4.2.3. Evaluation and analysis issues

Issue 6: Low statistical power

Scholars have long recognized that moderation tests typically have lower statistical power than main effect analyses due to factors such as variable intercorrelations, random effect designs, range restrictions, scale coarseness, non-normal outcome transformations, and differing variable distributions (Aguinis et al., 2017; Carte & Russell, 2003; Dawson, 2014; Memon et al., 2019). While Aguinis et al. (2017) documented that 43.41 % of studies failed to address statistical power considerations, our systematic review reveals marked improvement, with only approximately 15.33 % of contemporary studies neglecting this critical aspect. This progress likely stems from two key developments: (1) increased data accessibility through technological advancements enabling larger sample sizes, and (2) greater methodological awareness regarding scale coarseness and range restriction issues (incidence rates below 5 % in our review, see Table 1). Nevertheless, it is still essential to pay attention to low statistical power risk. Beyond the measurement reliability enhancements discussed in Issue 4, expanding sample sizes can effectively counteract power reduction (Memon et al., 2019). We strongly recommend conducting a priori power analysis using tools such as G* Power to determine the required sample size for detecting the hypothesized effect size for the proposed models (Faul et al., 2009; Sarstedt et al., 2023). Furthermore, regardless of whether moderation is supported, researchers should compute and report the observed statistical power. This practice allows readers to assess if non-significant results stem from low power or a truly insignificant relationship. A key metric indicating effect size of moderation analysis in multiple regression is the f^2, which quantifies the incremental explanatory power of the interaction term. It is calculated as follows:

$$f^2 = \frac{R_2^2 - R_1^2}{1 - R_2^2} \tag{6}$$

where R_1^2 is from Eq. 1 and R_2^2 is from Eq. 2. Conventional guidelines (Cohen, 1988) suggest that f^2 values of 0.02, 0.15, and 0.35 represent small, medium, and large effect sizes, respectively, in behavioral sciences. However, these are only benchmarks, and the appropriateness of an effect size should be judged within the specific research context.

Issue 7: Converting a continuous variable into categorical without appropriate justification

Researchers sometimes categorize continuous moderators into dichotomous variables to facilitate techniques like MGA (Aguinis, 2017; Becker et al., 2023; Carte & Russell, 2003; Dawson, 2014; Memon et al., 2019; Rasoolimanesh et al., 2021). Common approaches involve splitting a continuous moderator into "high" and "low" subgroups based on values above or below the mean or median, using k-means cluster analysis, or employing quartiles (Rasoolimanesh et al., 2021). Aguinis et al. (2017) report that approximately 10.24 % of articles employ this practice; our analysis similarly finds 10.95 % of studies do so. Actually, this dichotomization risks significant information loss, reducing statistical power and potentially altering the significance level of the product term (Cohen, 1983; Cohen et al., 2014; Irwin & McClelland, 2001). Furthermore, methods like median-splitting may create artificial subgroups lacking empirical or theoretical justification, raising questions about the rationale for specific split points (Dawson, 2014). Given these limitations, we advise researchers to avoid dichotomization. Instead, we recommend using SEM-based approaches to test the significance of the product term XM, as these methods retain the full information of the continuous moderator (Aguinis et al., 2017). If dichotomization is deemed necessary, researchers must provide compelling evidence justifying this choice and rigorously ensure measurement invariance across the resulting categories (see Issue 8 for details).

Issue 8: Lack of attention to measurement variance across categories

Measurement invariance is a vital prerequisite for guaranteeing the consistency and comparability of measurement tools across diverse groups or contexts in business and management research (Leitgöb et al., 2023). Establishing measurement invariance enhances both the reliability and generalizability of research findings. Our analysis revealed that 13.50 % of studies failed to address measurement invariance. While this percentage may appear relatively small, it is noteworthy that among the 274 reviewed articles, not all incorporated categorical variables. Therefore, this 13.50 % figure represents a significant methodological concern that merits attention. The examination of categorical moderators becomes particularly problematic under conditions of measurement non-equivalence. For example, in studies examining gender moderation, an unbalanced sample (e.g., 80 % female vs. 20 % male) may introduce variance bias that obscures true moderating effects. Such imbalance not only risks underestimating moderation but also confounds statistical significance with sample size disparities. To address these issues, researchers are advised to maintain balanced sample sizes across categorical groups whenever feasible. Furthermore, researchers should rigorously conduct measurement invariance tests prior to performing MGA. This ensures that measurement models produce equivalent representations across groups (Hair et al., 2022).

Issue 9: Opaque or incorrect interaction term creation with latent variables studies

The creation of interaction terms in latent variable moderation analysis continues to pose significant methodological challenges, particularly regarding transparency and accuracy (Becker et al., 2023; Rasoolimanesh et al., 2021). Empirical evidence reveals concerning reporting practices, with 73.47 % of CB-SEM studies and 50 % of PLS-SEM studies failing to adequately document their interaction term generation procedures (Rasoolimanesh et al., 2021). Our analysis corroborates these findings, identifying similar reporting deficiencies in 9.85 % of examined studies, which rises considerably if we only count studies with latent variables. This methodological opacity not only obscures potential flaws but also hinders study replication. A prevalent issue involves the manual computation of latent variable scores (e.g., through simple summation or averaging of indicators) followed by multiplication to create interaction terms - an approach that ignores differential indicator weighting and introduces measurement error bias (Becker et al., 2023; Rasoolimanesh et al., 2021). Rigorous methodological reporting and appropriate technique selection are therefore imperative for valid moderation analyses.

Actually, SEM offers three approaches for interaction term generation: (1) the product indicator method (Chin et al., 2003), (2) the orthogonalization approach (Little et al., 2006), and (3) the two-stage method (Hair et al., 2022). Each technique presents distinct advantages and limitations. The two-stage method demonstrates superior statistical power and represents the sole viable option for formative constructs, while the orthogonalization approach excels in point estimation and prediction accuracy (Henseler & Chin, 2010). Notably, both product indicator and two-stage methods inherently produce multicollinearity, whereas orthogonalization effectively mitigates this issue (Fassott et al., 2016). Method selection should consider both measurement characteristics (reflective vs. formative constructs) and research priorities (statistical power, estimation accuracy, or predictive performance) (Hair & Hult, Ringle, et al., 2022). For formative construct models, the two-stage method is unequivocally recommended (Henseler & Chin, 2010), while either two-stage or orthogonalization approaches are suitable for reflective constructs given PLS-SEM's robustness to collinearity (Becker et al., 2018).

Issue 10: Debate on mean-center/standardize X and M

The practice of mean-centering or standardizing variables in moderation analysis remains a subject of ongoing methodological debate. Our systematic review revealed that 42.70 % of studies explicitly reported performing such transformation prior to generating product terms, while the remainder did not report or use them. A key clarification is needed: mean-centering changes the scale of the variables but does not alter the underlying multicollinearity issue; it reduces non-essential collinearity, making coefficient estimates more stable, but the fundamental relationship remains (Aguinis et al., 2017; Becker et al., 2023; Dawson, 2014; Memon et al., 2019; Rasoolimanesh et al., 2021). Consequently, the core debate centers on whether the benefits for interpretation justify the procedure, given that unbiased estimation is possible with uncentered variables. The principal value of mean-centering lies in improving the interpretability of coefficients. When continuous variables are centered, the simple slope coefficients (b_1 and b_2) represent the effect of X when M is at its mean, and vice versa, which is often more meaningful than interpreting effects when X and M are zero. This aligns with recommendations to aid result interpretation (Aiken et al., 1991; Aguinis et al., 2017). For binary moderators, standardization is generally discouraged as it can obscure meaningful interpretation, since the reference point of a binary variable lacks substantive quantitative meaning (Becker et al., 2023). In such cases, analyzing unstandardized coefficients is preferable. We recommend that researchers consider mean-centering continuous predictors and moderators to enhance the interpretability of coefficients, as this practice facilitates a more meaningful understanding of simple slopes (Aguinis et al., 2017; Hair et al., 2022). In contrast, for binary moderators, using uncentered variables is generally advised to preserve clear and substantively meaningful interpretation (Becker et al., 2023). Regardless of whether centering is applied, it is essential to diagnose multicollinearity by reporting Variance Inflation Factors (VIFs) for all predictors, as VIFs (e.g., > 10) may indicate estimation issues that centering alone cannot resolve (Hair et al., 2009). Should severe multicollinearity persist, alternative techniques such as residual centering (orthogonalization approach) can be considered to separate the interaction effect from its constituent terms, though researchers should note the ongoing methodological debate regarding this approach. Ultimately, the decision to center should be driven by interpretive clarity and transparency, with researchers clearly reporting and justifying their chosen method.

Issue 11: Lack attention of lower-order interactions in three-way interaction model

In analyses involving multiple moderators, researchers frequently misinterpret three-way interactions (Becker et al., 2023). Although only 14 of 274 (5.11 %) reviewed articles exhibited this specific issue, its significance remains substantial—particularly given the relatively limited number of studies testing three-way interactions. A critical methodological error occurs when researchers test three-way interactions (XMW) without first including all constituent lower-order interaction terms (XM and XW) in the model. This omission is problematic because the three-way interaction effect is conditional upon these lower-order terms (Dawson, 2014). Failing to incorporate these necessary components can yield erroneous conclusions about both the three-way interaction and the overall model interpretation. As Becker et al. (2022) emphasize, researchers must rigorously specify models to include all relevant interaction terms. Utilizing appropriate statistical software (e.g., SmartPLS) that supports complex interaction estimation is equally essential. Adherence to these principles prevents common analytical pitfalls and ensures both validity and interpretability of findings in multi-moderator models.

4.2.4. *Interpretation and reporting issues*

Issue 12. Lack of attention of moderation effect size

The MMR method consistently identifies difference in R-squared values (ΔR^2) as the appropriate indicator for assessing moderation effect size (Chin et al., 2003; Hair et al., 2022). However, many articles neglect to report the moderation effect size or mistakenly use the interaction coefficient (b_3) (i.e., the coefficient of the interaction term) as the sole indicator of the moderation effect size (Becker et al., 2023; Carte and Russell; Rasoolimanesh et al., 2021). Our review found that 71.53 % of articles exhibit this issue, which poses significant risks to the validity and interpretability of the research findings. Indeed, test statistics regarding "H_0: $b_3 = 0$" and "H_0: $\Delta R^2 = 0$" consistently have identical mathematical conclusions, but ΔR^2 and b_3 are unequal unless the product term is measured without errors and the variances of Y and the product term are equal (Carte & Russell, 2003). Ignoring the moderation effect size can lead to misleading conclusions about the strength and importance of the moderator variable. For instance, a small but significant b_3 value may suggest a modest moderation effect, whereas the actual ΔR^2 might reveal a substantial increase in explained variance. Conversely, a large b_3 value might overstate the effect size if the product term explains a relatively small portion of the variance in the dependent variable. Researchers should quantify moderation effect size by comparing the model R-squared before and after adding the interaction term, not by relying on the value of b_3. A ΔR^2 around 0.02 is typically considered a small but meaningful effect, serving as a useful benchmark (Cohen et al., 2014). The f^2 statistic is also recommended as a complementary measure, as it directly reflects the unique contribution of the interaction term (Aiken et al., 1991). Crucially, no single metric provides a complete picture. The effect size (ΔR^2 or f^2) should be interpreted in conjunction with the significance of the interaction term (b_3) and, whenever possible, supported by a simple slope analysis plot to visualize the nature of the interaction. This multi-faceted approach—combining significance testing, effect size quantification, and graphical interpretation—ensures a rigorous and comprehensive assessment of moderation effects, enhancing the validity and practical relevance of research findings.

Issue 13. Ignoring nested model difference when interpreting main effect

A common practice in MMR is to first estimate a baseline model excluding the interaction term to interpret predictor main effects, subsequently adding the product term to test for moderation. Our review indicates that approximately 29.20 % of studies adopt this approach. When moderation exists, the X-Y relationship is not constant but conditional, varying systematically across levels of the moderator M (Aiken et al., 1991; Dawson, 2014). Critically, the main effect coefficient for X estimated in the baseline model (excluding the interaction) represents an average effect across levels of M. This practice risks misinterpretations—including Type I or Type II errors regarding the existence or strength of the X-Y association—particularly when main effect estimates differ significantly between nested models. Consequently, scholars advocate interpreting main effects based on the full model containing

Table 2
Guidelines for research involving moderation analysis.

Issues	Should do	Should not do
1.Theory development	• Determine whether to examine moderation and identify the moderators based on actual theoretical needs (Issue 1) • Theoretically develop the moderating hypotheses and specify how M impacts the relation of X and Y and the scope of the effect in advance (Issue 3) • Propose separate hypotheses for each moderator (Issue 14). • Theoretically justify the causal sequence X→Y and the moderation effects of the moderator (Issue 2) • Ensure the complex types of moderation models are used appropriately (Issue 2)	• Investigate moderation without theoretical support (Issue 1, 2) • Lack of the moderation hypotheses in advance (Issue 3) • Determine the ordering of variables based on a post hoc test (Issue 2)
2.Research design	• Estimate the needed sample size for the expected effect (Issue 6) • Use reliable measures, and the choice of scale should be guided by a thorough evaluation of several factors: research objectives, the potential for information loss, and statistical power required, characteristics of the respondents, and established norms and findings from previous studies (Issue 4) • Use measures having sufficient scale points and try to capture the full range of values for all variables if possible (Issue 6)	• Lack of attention to statistical power and required sample size (Issue 6) • Lack of attention range restriction when designing measures and scales (Issue 6)
3.Data collection	• Collect a large sample size that equals or exceeds the estimated number (Issue 6) • Conduct a pre-test and keep essential items up in order when the questionnaire is relatively lengthy to increase the reliability (Issue 4, 6) • Screen out suspicious responses (e.g., straight lining and zigzag patterns) and responses that have less than 0.5 SD due to small variation (Issue 6)	• Collect small sample size (Issue 6)
4.Data analysis	• Conduct another power analysis to determine the statistical power of the collected data (Issue 6) • Select the appropriate approach and tool: e.g., PLS-SEM is preferable for latent variable models • Eliminate the curvilinear effects before testing the moderation if X and M are highly correlated • If the moderators are continuous variables: • Mean-center/z-standardize the variables before creating the product term. (Issue 10) • If severe multicollinearity persists, use the residual centering approach such as orthogonalization (Issue 10) • Select an appropriate method to generate the product term: e.g., an orthogonalizing method for CB-SEM/PLS-SEM with reflective constructs; a two-stage method for PLS-SEM with formative and reflective constructs (Issue 9) • Ensure the product terms and all of its components are in the final regression model (Issue 13, 14) • Test the coefficients of each product term if there are multiple predictors • Use bootstrap estimates if the variables do not meet the parametric assumptions (Issue 5) • Convert continuous moderators into categorical with appropriate justification to use MGA (Issue 7) • If the moderators are categorical variables: • Suggest using MGA and examine measurement invariance across categories (Issue 8) • Eliminate the interactions between moderators if there are multiple moderators in the model	• Artificially median split the continuous moderators into dichotomous variables without appropriate justification (Issue 7) • Use MGA without a measurement invariance test (Issue 8)
5.Results interpretation	• Interpret the significance of moderation effects based on the effect size ΔR^2 or f^2 (Issue 12) • Interpret the main effect and moderation in the full model (Issue 13, 14) • Interpret how M affects the relationship between X and Y by visualizing and probing moderation (Issue 15)	• Use b_3 to represent the moderation effect size (Issue 12) • Interpret the main effect and moderation in the partial model (Issue 13, 14) • Lack of attention to moderation probe (Issue 15)
6.Result reporting	• Report scale reliabilities (Issue 4) • Report measurement invariance test results for categorical moderator (Issue 8) • Report transformations with theoretical rationale (Issue 5) • Report statistical power (Issue 6)	• Lack of scale reliability report (Issue 4) • Lack of measurement invariance test (Issue 8) • Lack of statistical power report (Issue 6, 12)

the interaction term (Aguinis et al., 2017; Becker et al., 2023; Memon et al., 2019; Rasoolimanesh et al., 2021). We recommend retaining the sequential approach but reporting both sets of estimates with explicit model specifications to highlight potential discrepancies. Where significant changes in main effect estimates occur, researchers should conduct simple slope analyses (see Issue 15) to probe the interaction's nature and reveal the conditions under which X affects Y across M's range.

Issue 14. Interpreting moderation based on the moderated model excluding X and M

Early methodological work addressing multicollinearity concerns in moderation analysis proposed omitting the components (X and M) and testing reduced models containing only the product term (Y = XM; Sharma et al., 1981). Some researchers adopted this approach when it appeared to enhance statistical significance, a practice that persists in approximately 26.64 % of contemporary studies - an improvement from the 43.9 % prevalence reported by Aguinis et al. (2017), yet still methodologically problematic. This specification suffers from fundamental limitations (Becker et al., 2023; Dawson, 2014). The Y-XM correlation inherently depends on the constituent variables' properties (Irwin & McClelland, 2001), potentially conflating main and interaction effects. This confounding becomes particularly acute when main and moderation effects share directional signs (Cohen, 1978), compromising the reliability of moderation tests. Consequently, scholars must include all constituent terms alongside their product in moderation analyses regardless of their statistical significance (Cronbach, 1987), as proper model specification supersedes collinearity concerns.

Issue 15. Lacking attention to the moderation probe

Many existing moderation studies lack the necessary attention to moderation probes (51.83 % of our examined articles), although probing the conditional effect of X on Y at a specific value of M can provide informative messages, regardless of whether a moderation test is significant (Hayes, 2018). Given that the significance of the product term does not indicate how M affects the direction and magnitude of the association, visualizing and probing moderation with appropriate methods is necessary. Two primary methods exist to probe an interaction: the pick-a-point and the J–N approach, which we have elaborated on in session 4.3. Both approaches are available in regression-based and SEM-based software (e.g., PROCESS macro, SmartPLS). We strongly recommend reading Hayes (2018) and Ringle et al. (2015) to understand better how to perform these two approaches in PLS-SEM. Additionally, Murphy and Aguinis (2022) provide a guide for improving the visualization and reports of moderation effects.

5. Stage-based state-of-the-art recommendations

Building on this diagnostic foundation, we operationalize the solutions through Mertens and Recker's (2020) six-stage research framework. Table 2 presents stage-based "should do/should not do" recommendations that address Bergh et al. (2022) call for more tailored and specific step-by-step guidance to gauge the choices in moderation research. The proposed recommendations not only resolve cross-stage issues (e.g., Issue 6) but also enables researchers to systematically optimize the validity, efficiency, and robustness of their moderation analysis.

For researchers who begin a new moderation study, they should first develop hypotheses based on theories. The primary task of the theory development stage is to theorize the moderating relationship. Researchers' activities in this stage involve preparing the research model and constructing the hypotheses. Researchers need to determine whether to examine moderation and identify the potential moderators based on the requirements of the focal research question and the support from theory and literature. The causal relation between X and Y and why M is the moderator and X is the predictor should be clearly and theoretically proposed. In particular, researchers should specify how the moderator affects the relation and the scope of the effect in advance. If possible, we encourage researchers to explore the theoretically driven complex models (e.g., multilevel models) with appropriate methods.

Once the model and hypotheses are established, researchers must design a plan to collect data (e.g., determine the sample size, measures, scale, and sample). To achieve acceptable statistical power, the researchers should estimate the required sample size based on an expected effect size. In survey research, the choice of scale should be guided by a thorough evaluation of several factors, including research objectives, potential for information loss and statistical power required, characteristics of the respondents, and established norms and findings from previous studies. In particular, scales capturing the full range of values for all variables are preferable, and a ratio scale is recommended compared with an interval scale for the continuous predictors and moderators. Based on the research design, researchers are expected to collect relatively large-sized data from the planned sample. To ensure reliability, researchers should conduct a pre-test and keep essential items in order when the questionnaire is lengthy. Additionally, researchers can consider screening out suspicious responses (e.g., straight-lining and zigzag patterns) and responses with less than 0.5 standard deviance due to small variation.

In the data analysis stage, researchers need to perform another power analysis to check the statistical power of the gathered data before selecting an appropriate approach and tool for verifying the hypotheses. If X and M are highly correlated, researchers should eliminate the curvilinear effects before testing the moderation effects. Researchers are also suggested to examine VIFs to assess and address potential multicollinearity among variables. If severe multicollinearity persists despite these precautions, residual centering approach such as orthogonalization can be considered to separate the interaction effect from its constituent terms If the moderators are categorical, MGA with measurement invariance test is better for researchers. If a moderator is a continuous variable, researchers can either test it using the product term or convert it into a categorial variable with appropriate justification. To test the product term, researchers are recommended to mean-centre/z-standardise the predictors and moderators before generating the product terms. Regression-based or PLS-SEM is advised to test the significance of the product terms. If the PLS-SEM is employed, the orthogonalizing method is recommended for a model with reflective constructs to generate the product term. In contrast, the two-stage method is recommended for formative and reflective constructs. The moderated model should include all independent variables, moderators, and product terms. A bootstrap estimate is recommended if the variables do not satisfy the normal distribution assumptions. In addition, the researchers should eliminate the interactions between moderators if multiple moderators exist in the model.

Researchers then interpret the results based on the statistical tests. Great care should be taken when concluding the moderation effect size. Conclusions should derive from comparing the main-effects-only model with the full model that includes the interaction term, its components (X and M), and relevant covariates. Moderation effects should be evaluated using effect size metrics (ΔR^2 or f^2) rather than relying solely on the coefficient b_3. We also recommend probing significant interactions using established techniques to illustrate their nature. This probing process also helps delineate the conditional nature of main effects across the range of M.

Finally, the researchers are suggested to report the scale reliabilities and validities for the predictors, moderators, and product terms; measurement invariance test results for categorical moderators; theoretical rationale for all transformations; and statistical power of the gathered data.

6. Conclusions

Moderation analysis plays an increasingly pivotal role in empirical research within the fields of IS and other business and management domains. Despite numerous discussions on the issues within moderation analysis, errors and problems persist in moderation research, which hinders researchers from harnessing the full potential of moderation analysis. As our comprehension of the nuances of moderation analysis deepens (as evidenced by studies such as Aguinis et al., 2017; Montoya, 2019; Rasoolimanesh et al., 2021; Becker et al., 2023), continuous synthesis of latest findings is essential to identify challenges and improve moderation research. Therefore, this research provides a meticulous examination of historical and theoretical underpinnings through a rigorous review of 30 seminal works. By synthesizing the most current scholarly insights, our study not only traces the evolution of moderation analysis but also enhances the understanding of various moderation models and their conceptual distinctions. We have critically assessed the utility of different moderation models and their analytical methods, providing tailored guidance for researchers to select the most suitable options. Moreover, our study highlights existing gaps and proposes practical resolutions by identifying and addressing 15 prevalent issues with a systematic review of 274 moderation articles published in top-tier IS journals. A phased guideline covering the entire research cycle, from planning to analysis and reporting, is vital for researchers. It provides a structured framework for moderation analysis and demonstrates practical implementation steps. By offering a novel theoretical framework for empirical application, this study advances both theoretical and practical progress in moderation analysis.

6.1. Theoretical and methodological implications

While these prior works systematically documented methodological challenges, they lack contemporary empirical evidence and a cohesive, lifecycle-spanning organizational framework (e.g., Aguinis et al., 2017; Carte & Russell, 2003; Memon et al., 2019; Rasoolimanesh et al., 2021). Our research directly fills the gap through four key contributions. First, building upon the historical groundwork, we synthesize 30 seminal publications to establish a robust theoretical foundation and trace the conceptual evolution of moderation analysis, enabling researchers to efficiently grasp its developmental trajectory. Crucially, we extend this historical perspective with a systematic review of 274 recent articles (2022–2024), providing current empirical evidence that quantifies the persistence of 15 historical issues (e.g., a 72 % prevalence of b_3 misinterpretation as effect size) and identifies 5 previously common problems that have significantly diminished.

Second, another significant contribution is the development of an integrated, stage-based best practices framework (Table 2). This framework transcends the limitations of prior guidelines, which often focused on specific aspects (e.g., analysis, interpretation) or offered general recommendations. Instead, it provides actionable, empirically grounded guidance systematically organized across the complete research lifecycle – from theory development and preparation to result interpretation and final reporting – directly addressing the call by Bergh et al. (2022) for phase-specific methodological support.

Third, through an extensive examination of the literature, we clarify the conceptual intricacies of major moderation models (i.e., two-way, three-way interactions, and moderated mediation) and systematically classify them alongside their corresponding analytical approaches. This synthesis facilitates enhanced researcher understanding of moderation effects and assists in selecting the most appropriate methods for given contexts.

Finally, extending current literature comprehensively, we identify, elaborate on, and diagnose the current status of 15 critical issues throughout the research process. For each problem, we provide empirically validated assessments and evidence-based solutions contextualized within our stage-based framework. This not only highlights potential pitfalls in existing research but offers concrete strategies to avoid them. Furthermore, Appendix B presents an application case using PLS-SEM and CB-SEM, demonstrating the framework's practical implementation and offering a template for applying theoretical guidance, a contribution previously absent in the literature.

6.2. Limitations and future research

Although this study aims to provide comprehensive research-stage-based recommendations on moderation analysis, other factors must exist, but we did not discuss and/or include in the guidelines that also influence moderation research. Future research is thus also encouraged to review and assess moderation research's evolution continuously and continually guide future research by focusing on the differences in theory development, research design, data collection and analysis, interpretation, and reporting.

CRediT authorship contribution statement

XU Yujing: Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wen-Lung Shiau:** Writing – review & editing, Supervision, Software, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijinfomgt.2025.102995.

Data availability

Data will be made available on request.

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